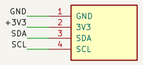
HastaDAQ

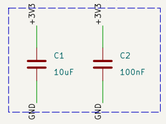
Net labels connect everything: same name = same net. +3V3 / GND are plain labels, not power symbols.

Pimoroni Pico Plus 2 (+ RM2 on SP/CE): same 40-pin header as Pico 2 W. I2C0 = GP4/GP5 (header pins 6/7), hard-wired to the IMU Qwiic-breakout header and both ADS122C04 boards by header pins. Crossed-out pins are intentionally unused.

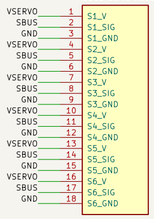
J1
IMU via Qwiic breakout (The Pi Hut 106676)



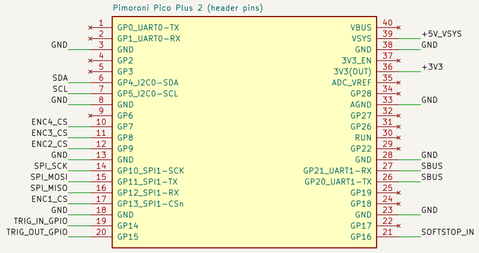
ADS1 supply decoupling (place at ADC)



J8
SBUS servo header 3x6 (6 x V, SIG, GND)



U1
Pimoroni Pico Plus 2 (header pins)



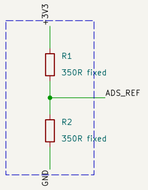
U2
Adafruit ADS122C04 #1 (I2C 0x40)



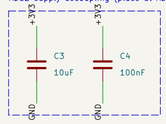
J2
HB1 half-bridge #1 (on beam)



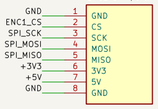
Shared completion resistors (fixed) -> A1 of BOTH ADCs



ADS2 supply decoupling (place at ADC)



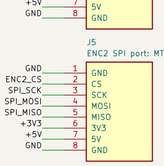
J4
ENC1 SPI port: MT6835 (pitch) / Orbis



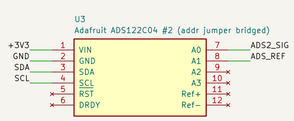
J3
HB2 half-bridge #2 (on beam)



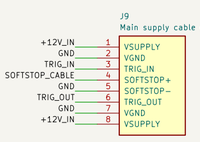
J5
ENC2 SPI port: MT6835 / Orbis



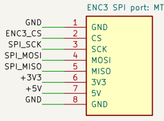
U3
Adafruit ADS122C04 #2 (addr jumper bridged)



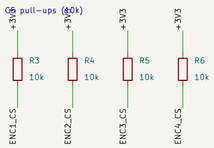
J9
Main supply cable (2x4): VSUPPLY, VGND, TRIG, SOFTSTOP



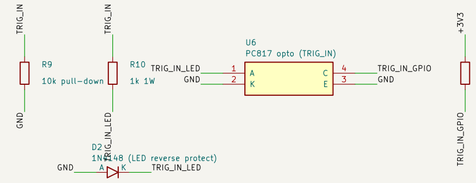
J6
ENC3 SPI port: MT6835 / Orbis



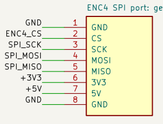
CS pull-ups (10k)



TRIGGER IN: any 3.3 V .. 24 V level, opto input



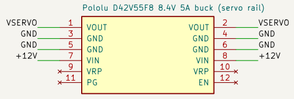
J7
ENC4 SPI port: generic SPI device



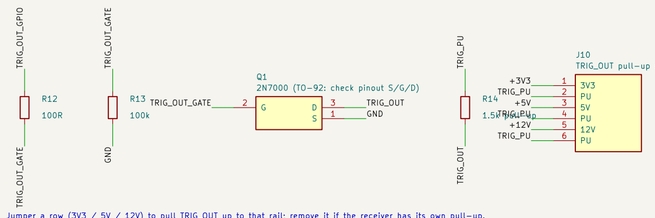
SOFTSTOP_CABLE
R7
1k (soft-stop series)
SOFTSTOP_IN

+3V3
R8
10k (soft-stop pull-up)
SOFTSTOP_IN

U4
Pololu D42V55F8 8.4V 5A buck (servo rail)



TRIGGER OUT: open-drain, pull-up rail chosen by jumper



F1
Fuse (size to servo load)
+12V_IN
+12V

+12V
C5
100uF input bulk (>=25V)
VSERVO_GND
C6
470–1000uF servo bulk (>=16V)
GND

U5
SparkFun BabyBuck 5V (AP63357) (Pico VSYS, Orbis)



D1
Schottky (e.g. 1N5822, 3A)
+5V
A
K
+5V_VSYS

Jumper a row (3V3 / 5V / 12V) to pull TRIG_OUT up to that rail: remove it if the receiver has its own pull-up.
Pull-up current: 3.3 V ~2 mA, 5 V ~3 mA, 12 V ~7 mA (enough for a 6.5 mA opto input such as MPS4200).
24 V receivers: no jumper, add an external 24 V pull-up in the harness. 3V3 for eSync 2 high-Z ports (3.3 V max).

NOTES
1. ONE common GND (Pico, sensors, servo PSU). SBUS needs a shared ground reference.
   RM2 WiFi breakout (off-board, cabled to the Pico Plus 2 SP/CE connector): uses GP32–GP36 only, no overlap with I2C (GP4/5) or any other signal.
2. ADS2: bridge one address jumper on its back (A0/A1); resulting address unconfirmed — firmware falls back to an I2C scan at boot. I2C pull-ups: rely on the Adafruit breakouts' on-board pull-ups.
3. Each half-bridge module (2 gauges) plugs into a 3-pin header: V, SIG (mid-point), GND. SIG -> A0 of its ADC, direct (no input filter). ONE shared pair of fixed 350R completion resistors: its mid-point (ADS_REF) goes to A1 of BOTH ADCs. Excitation is the ADC's own VIN, so VREF_SUPPLY is ratiometric — no Ref+/Ref- wiring.
4. SBUS: GP20 (TX) and GP21 (RX) on one net (UART inverted in firmware). One 3x6 servo header (6 plugs of V, SIG, GND; footprint HastaDAQ:PinHeader_3x06): V = VSERVO (8.4 V), SIG = SBUS (all six in parallel), GND.
5. POWER: 12 V in -> fuse -> +12V rail -> (a) Pololu D42V55F8 = VSERVO 8.4 V (servos), (b) SparkFun BabyBuck 5V = +5V (Orbis, Schottky -> Pico VSYS). VSYS max is 5.5 V. USB (VBUS) is left for programming; with USB only, +5V is not live.
6. Trigger pins are 3.3 V logic only. Pin order on the headers: see each symbol.
   Main supply cable (2x4, pins 1–8 row-major): 1,8 VSUPPLY (12 V) | 2,7 VGND | 3 TRIG_IN | 6 TRIG_OUT | 4 SOFTSTOP+ | 5 SOFTSTOP-.
   SOFTSTOP+ -> 1k -> GP16 (10k pull-up to 3V3). SOFTSTOP- = GND: loop closed = run, loop broken = stop (fail-safe on a cable break).
   Use a keyed/shrouded header: a 0.1in header pin is only good for ~1–3 A, so check the servo peak current against 2 supply pins.
7. Encoders: 4 x 2x4 headers (pins 1–8: GND, CS, SCK, MOSI, MISO, 3V3, 5V, GND) on one SPI1 bus; CS = GP13/GP9/GP8/GP7.
   MT6835 uses 3V3 (3.3 V logic); RLS Orbis uses 5V with 3.3 V logic. Check MISO tri-states when CS is high.
   The four ENC headers are a generic SPI port (SPI1: SCK GP10, MOSI GP11, MISO GP12, own CS each, 3V3 and 5V supply), so any 3.3 V SPI device can be plugged in, not just encoders. Each device needs its own CS and an MISO that tri-states when CS is high.
8. Use shielded twisted cable for the gauge wires, shield to GND at the ADC end only.

HastaArduino	
Sheet: /	
File: hastadaq.kicad_sch	
Title: HastaDAQ	
Size: A2	Date:
Rev: A	
KiCad E.D.A. 10.0.7	Id: 1/1